A PROJECT REPORT
ON

LIBRARY MANAGEMENT SYSTEM

Submitted to

The Department of Computer Science

In Partial fulfilment of the requirements for the
3rd Semester Project of

MASTER OF SCIENCE INFORMATION TECHNOLOGY (MSc IT)

Academic Year: 2026-2027

Silver Oak College of Computer Application
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CERTIFICATE

Date:

This is to certify that the project entitled "**LIBRARY MANAGEMENT SYSTEM**" has been carried out by **Student Name (Enrol. No.)** under the guidance of **Project Guide Name** in fulfillment of the 3rd Semester **Software Project II (IOM6043C)** at Silver Oak College of Computer Application, Silver Oak University, Ahmedabad during academic year 2026-2027.

Internal Guide
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Head of Department
Department of Computer Science

STUDENT DECLARATION

I hereby declare that the project entitled "LIBRARY MANAGEMENT SYSTEM" submitted in partial fulfilment of the requirements for the 3rd Semester of Master Of Science Information Technology (MSc IT) is an original work carried out by me under the guidance of Guide Name. I further declare that this project report, or any part thereof, has not been submitted by me or any other person for the award of any other degree or diploma of this or any other university.

I also declare that all the information furnished in this project report is based on my own study, analysis, and implementation, and due acknowledgement has been made wherever the work of others has been referred to.

Place: Ahmedabad
Date: _______________

Signature of the Student
Student Name
Enrollment No.

ACKNOWLEDGMENT

The development of the Library Management System was a collaborative undertaking, and I am deeply grateful to all who contributed to its successful completion.

I extend my sincere appreciation to Project_Guide_Name for his/her invaluable guidance, support, and expertise throughout this project. His/Her insightful feedback and direction were instrumental in shaping the system's design and functionality.

I am also grateful to the Department of Computer Science, Silver Oak College of Computer Application, for providing the essential resources and infrastructure that made the development of the Library Management System possible. Their support was invaluable.

Furthermore, I would like to thank the library staff and student members who generously participated in the requirement-gathering and testing phases. Their feedback was crucial in identifying areas for improvement and ensuring the system effectively addresses the real-world demands of a library environment.

This project would not have been possible without the collective efforts of everyone involved. I am truly grateful for their contributions and unwavering support.

Student Name

Enrollment No.

ABSTRACT

The Library Management System is a computerized application developed to automate library operations and replace traditional manual record-keeping with an efficient and user-friendly solution. The system manages book cataloguing, member registration, book issue and return, fine calculation, and report generation using a centralized database.

Developed using a modular approach, the system includes separate modules for book management, member management, and transaction management. It provides role-based access for administrators and library staff, ensuring secure management of library records. Advanced search and filtering features enable users to quickly locate books by title, author, category, or ISBN.

The primary objective of the project is to reduce manual effort, minimize human errors, and provide accurate and timely access to library information. The system also generates reports such as overdue books, frequently issued books, and fine collection summaries to support effective library administration. The project follows the Software Development Life Cycle (SDLC), including requirement analysis, system design, implementation, testing, and deployment.

The developed system improves the efficiency, accuracy, and reliability of library operations while enhancing the overall user experience. It also provides a scalable foundation for future enhancements.

Keywords: Library Management System, Automation, Database Management System (DBMS), SDLC, Web Application.

(Abstract will be different for each students as per their title selected)

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
SDLC	Software Development Life Cycle
DFD	Data Flow Diagram
ER Diagram	Entity-Relationship Diagram
UML	Unified Modeling Language
GUI	Graphical User Interface
SQL	Structured Query Language
IDE	Integrated Development Environment
ISBN	International Standard Book Number
OS	Operating System
RAM	Random Access Memory
UI/UX	User Interface / User Experience
CRUD	Create, Read, Update, Delete

ACHAPTER 1

Introduction

1.1 Background

Libraries have traditionally served as centres of knowledge, providing access to books, journals, and other reference material to students, faculty, and the general public. However, the manual processes historically used to manage library operations — such as maintaining registers for book records, issuing and returning books, and calculating fines — are time-consuming, prone to human error, and difficult to scale as the volume of books and members grows.

With the rapid advancement of information technology, it has become both feasible and necessary to computerize such repetitive and record-intensive tasks. A Library Management System (LMS) automates the core functions of a library, including cataloguing, circulation (issue/return), member management, and report generation, thereby improving efficiency, accuracy, and accessibility of information for both library staff and members.

1.2 Problem Statement

The existing manual library system at many institutions relies on physical registers and spreadsheets to track books and members. This approach suffers from several drawbacks: difficulty in searching for a specific book, delays in checking book availability, inaccurate fine calculation, lack of real-time reporting, and the risk of losing records due to physical damage. There is, therefore, a need for a centralized, computerized system that can manage all library operations efficiently, accurately, and securely.

1.3 Objectives of the Project

- To design and develop a centralized database for storing book and member information.
- To automate the process of book issue, return, and fine calculation.
- To provide an efficient search mechanism for locating books by title, author, category, or ISBN.

- To generate reports such as overdue books, issued books, and fine collection summaries.
- To provide role-based access control for administrators and library staff.
- To reduce manual effort, paperwork, and human error in library operations.

1.4 Scope of the Project

The scope of this project is limited to automating the core functional areas of a college or departmental library, namely book cataloguing, member registration, issue/return transactions, fine management, and basic reporting. The system is designed for use by library administrators and staff; it does not, in its current version, include an online public access catalogue (OPAC) for external users or integration with third-party payment gateways, both of which are identified as areas for future enhancement.

1.5 Organization of the Report

This report is organized into seven chapters. Chapter 1 introduces the project background, problem statement, objectives, and scope. Chapter 2 reviews existing library management solutions and related literature. Chapter 3 details the system requirement analysis, including functional and non-functional requirements. Chapter 4 describes the system design and methodology, including architecture, data flow, and database design. Chapter 5 discusses the implementation details and coding approach. Chapter 6 presents the testing strategy, test cases, and results analysis. Finally, Chapter 7 concludes the report and outlines directions for future work.

CHAPTER 2

Literature Review

2.1 Existing Systems

Several commercial and open-source library management solutions are currently available, ranging from simple spreadsheet-based tracking tools to comprehensive enterprise systems such as Koha and SirsiDynix Symphony. These systems typically provide cataloguing, circulation, and reporting modules; however, many are either too expensive for small institutional libraries or too generic to be tailored to the specific workflow of a departmental library.

A review of academic literature on library automation indicates that computerized systems significantly reduce the average time required for book issue and return transactions, improve the accuracy of fine calculation, and provide better visibility into inventory through real-time reporting, when compared with manual record-keeping systems.

2.2 Comparative Analysis

A comparative study of existing systems reveals common strengths such as barcode-based cataloguing, automated fine calculation, and multi-user access. Common limitations include high licensing costs, complex deployment requirements, and limited customization options for smaller institutions. The proposed Library Management System aims to combine the essential features of these existing solutions into a lightweight, cost-effective, and customizable application suited to a college-level library.

2.3 Summary

The literature review establishes a strong case for automating library operations and highlights the specific gaps — namely cost, complexity, and lack of customization — that the proposed system seeks to address through a modular, database-driven design.

CHAPTER 3

System Requirement Analysis

3.1 Functional Requirements

Functional requirements define the specific behaviours and functions that the system must support. Table 3.1 summarizes the key functional requirements identified for the Library Management System.

Req. ID	Requirement Description
FR-1	The system shall allow administrators to add, update, and delete book records.
FR-2	The system shall allow registration and management of library members.
FR-3	The system shall allow issuing and returning of books against a member ID.
FR-4	The system shall automatically calculate fines for overdue books.
FR-5	The system shall allow searching of books by title, author, category, or ISBN.
FR-6	The system shall generate reports of issued books, overdue books, and fines.
FR-7	The system shall provide role-based login for Admin and Staff users.

3.2 Non-Functional Requirements

- Usability: The system shall provide a simple and intuitive graphical user interface.
- Reliability: The system shall ensure accurate and consistent data storage and retrieval.
- Performance: Search and transaction operations shall complete within two seconds under normal load.
- Security: The system shall restrict access to authorized users through login credentials.
- Maintainability: The modular design shall allow easy addition of new features in the future.

3.3 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above

Component	Specification
RAM	4 GB or above
Hard Disk	250 GB or above
Monitor	Standard colour monitor

3.4 Software Requirements

Component	Specification
Operating System	Windows 10 / 11
Front-End	HTML, CSS, JavaScript
Back-End	Python / Java / PHP (as applicable)
Database	MySQL
IDE	Visual Studio Code / NetBeans

3.5 Feasibility Study

3.5.1 Technical Feasibility

The technologies required for developing the proposed system — including the chosen programming language, database management system, and development tools — are freely available, well-documented, and within the technical capability of the development team.

3.5.2 Economic Feasibility

The project relies primarily on open-source and freely available tools, resulting in negligible development cost. The long-term benefits, in terms of reduced manual labour and improved accuracy, significantly outweigh the minimal implementation cost.

3.5.3 Operational Feasibility

The proposed system is designed with a simple and intuitive interface, requiring minimal training for library staff to operate effectively, thereby ensuring high operational feasibility.

CHAPTER 4

System Design and Methodology

4.1 System Architecture

The Library Management System follows a three-tier architecture consisting of a presentation layer (user interface), an application layer (business logic), and a data layer (database). This separation of concerns improves maintainability and allows each layer to be modified independently without affecting the others.

[Figure 4.1: System Architecture Diagram — to be inserted here]

4.2 Data Flow Diagrams

The Data Flow Diagram (DFD) illustrates the flow of information through the system. The Level 0 DFD (context diagram) represents the system as a single process interacting with external entities such as Admin, Staff, and Member. The Level 1 DFD decomposes this into major sub-processes, including Book Management, Member Management, and Transaction Management.

[Figure 4.2: Data Flow Diagram (Level 0) — to be inserted here]

[Figure 4.3: Data Flow Diagram (Level 1) — to be inserted here]

4.3 Entity-Relationship Diagram

The Entity-Relationship (ER) Diagram models the relationships among the key entities of the system, namely Book, Member, Transaction, and Staff. Each Book can be involved in multiple Transactions, and each Member can perform multiple Transactions, establishing a many-to-many relationship resolved through the Transaction entity.

[Figure 4.4: Entity-Relationship Diagram — to be inserted here]

4.4 Database Design

The database for the Library Management System consists of the following major tables, described below.

4.4.1 tbl_Book

Field Name	Data Type	Description
BookID	Int (PK)	Unique identifier for each book
Title	Varchar(150)	Title of the book
Author	Varchar(100)	Author name
ISBN	Varchar(20)	International Standard Book Number

4.4.2 tbl_Member

Field Name	Data Type	Description
MemberID	Int (PK)	Unique identifier for each member
Name	Varchar(100)	Member's full name
Contact	Varchar(15)	Contact number

4.4.3 tbl_Transaction

Field Name	Data Type	Description
TransactionID	Int (PK)	Unique identifier for each transaction
BookID	Int (FK)	Reference to tbl_Book
MemberID	Int (FK)	Reference to tbl_Member

4.5 UML Diagrams

Unified Modeling Language (UML) diagrams were used to model the system from different perspectives. The Use Case Diagram represents the interactions between actors (Admin, Staff, Member) and the system. The Class Diagram represents the static structure of the system in terms of classes, attributes, and methods. The Sequence Diagram illustrates the interaction sequence for the book issue process.

[Figure 3.1: Use Case Diagram — to be inserted here]

[Figure 4.5: Class Diagram — to be inserted here]

[Figure 4.6: Sequence Diagram — Book Issue Process — to be inserted here]

CHAPTER 5

Implementation and Coding

5.1 Development Environment

The Library Management System was developed using a standard integrated development environment configured with the front-end, back-end, and database technologies listed in Section 3.4. Version control was maintained throughout development to track changes and manage revisions of the source code.

5.2 Module Description

5.2.1 Login Module

This module authenticates Admin and Staff users based on a username and password, and redirects them to the appropriate dashboard based on their assigned role.

5.2.2 Book Management Module

This module allows authorized users to add, update, delete, and search book records stored in the database, including details such as title, author, ISBN, category, and available quantity.

5.2.3 Member Management Module

This module handles the registration, updating, and deletion of member records, along with viewing a member's active and historical transactions.

5.2.4 Transaction Module

This module manages the issuing and returning of books, automatically calculates due dates, and computes applicable fines for overdue returns based on a configurable per-day fine rate.

5.2.5 Report Module

This module generates reports such as currently issued books, overdue books, and fine collection summaries, which can be viewed or exported by the administrator.

5.3 Coding Standards

Consistent naming conventions, indentation, and inline comments were followed throughout the codebase to improve readability and maintainability. Reusable functions were used wherever possible to minimize code duplication, and input validation was implemented at both the client and server sides to maintain data integrity.

5.4 Sample Code Snippets

The following illustrative snippet demonstrates the fine calculation logic used in the Transaction Module (sample pseudocode for reference purposes):

```
IF ReturnDate > DueDate THEN     OverdueDays = ReturnDate - DueDate
Fine = OverdueDays * FineRatePerDay ELSE     Fine = 0 END IF
```

CHAPTER 6

Testing and Results Analysis

6.1 Testing Objectives

The primary objective of testing was to verify that each module of the Library Management System functions correctly, individually and in integration with other modules, and that the system meets the functional and non-functional requirements identified in Chapter 3.

6.2 Testing Methods

A combination of Unit Testing, Integration Testing, and User Acceptance Testing (UAT) was performed. Unit testing verified individual functions such as fine calculation and search filtering. Integration testing verified the correct interaction between the Book, Member, and Transaction modules. User acceptance testing was conducted with sample library staff to validate real-world usability.

6.3 Test Cases

6.3.1 Login Module

Test Case ID	Description	Expected Result	Status
TC-01	Login with valid credentials	User redirected to dashboard	Pass
TC-02	Login with invalid password	Error message displayed	Pass

6.3.2 Book Issue Module

Test Case ID	Description	Expected Result	Status
TC-04	Issue an available book	Book issued and quantity decremented	Pass
TC-05	Issue an already fully-issued book	Error message: no copies available	Pass

6.3.3 Book Return Module

Test Case ID	Description	Expected Result	Status
TC-07	Return book before due date	Book returned, no fine charged	Pass

Test Case ID	Description	Expected Result	Status
TC-08	Return book after due date	Fine calculated correctly	Pass

6.4 Results Analysis

All executed test cases across the Login, Book Issue, and Book Return modules produced the expected results, indicating that the core functionality of the Library Management System operates correctly and reliably. Minor issues identified during initial testing, such as incorrect fine rounding, were resolved prior to final deployment. Overall, the results confirm that the system meets the functional requirements defined in Chapter 3.

CHAPTER 7

Conclusion and Future Scope

7.1 Conclusion

The Library Management System successfully automates the core operations of a college library, including book cataloguing, member management, issue/return transactions, and fine calculation. The system significantly reduces the manual effort involved in maintaining library records, minimizes human error, and provides quick and accurate access to information for both library staff and administrators. Testing confirmed that the system meets its defined functional and non-functional requirements, demonstrating the practical benefits of automation over traditional manual record-keeping.

7.2 Future Scope

- Integration of an Online Public Access Catalogue (OPAC) to allow members to search the catalogue remotely.
- Implementation of a barcode/RFID-based system for faster book issue and return.
- Integration of an online payment gateway for fine payment.
- Development of a mobile application for members to check due dates and renew books.
- Addition of an email/SMS notification module to remind members of upcoming due dates.

BIBLIOGRAPHY & REFERENCES

[1] Author Name, "Title of Referenced Book," Publisher, Edition, Year.

[2] Author Name, "Title of Referenced Research Paper," Journal/Conference Name, Volume, Issue, Year, Page Numbers.

[3] Official documentation of the programming language / framework used (e.g., MySQL Reference Manual, https://dev.mysql.com/doc/).

[4] Official documentation of the development tools used (e.g., Visual Studio Code Documentation).

[5] Relevant online tutorials, technical articles, or W3Schools references consulted during development.

Note to student: Replace the above placeholder entries with the actual books, research papers, and websites referred to during your project, formatted consistently in IEEE or APA style as instructed by your guide.

APPENDICES

Appendix A Review 1 — Scanned Copy

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Appendix B Review 2 — Scanned Copy

[Insert scanned copy of the signed Review 2 evaluation sheet here.]

Appendix C Review 3 — Scanned Copy

[Insert scanned copy of the signed Review 3 evaluation sheet here.]

Appendix D Review 4 — Scanned Copy

[Insert scanned copy of the signed Review 4 evaluation sheet here.]

PUBLICATION / PATENT / RESEARCH OUTPUT

(This section is optional and applicable only if the student has published a research paper, filed a patent, or produced any other research output based on this project.)

[Insert scanned copy of the publication certificate / acceptance letter / patent filing receipt here.]